Bayesian Optimization driven Random Forest Classification for Efficient Crop Recommendation System

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Abstract

Agriculture is the backbone of India's economy; it provides livelihoods to millions and significantly contributing to the country's gross domestic product (GDP). As agriculture is the main pillar of india so selecting the right crop is very much essential for increasing productivity, conserving resources. A crop recommendation system helps farmers to make best decisions by analysing soil quality, weather conditions, and environmental factors. It helps farmers grow more crops by choosing the best ones for their soil and weather conditions, leading to better harvests and higher profits. By analysing important factors like soil nutrients, water levels, and weather changes, the system helps farmers to use water, fertilizers, and other resources wisely. It also helps them deal with changing weather by suggesting crops that can survive in different temperatures, rainfall, and extreme conditions. This reduces the chances of crop failure and supports stable farming and the system encourages eco-friendly farming, keeps the soil healthy, and prevents waste, making agriculture more sustainable for the future. The proposed model evaluates key factors such as Nitrogen (N), Phosphorus (P), Potassium (K), Humidity (H), Temperature (T), potential of hydrogen (pH) levels, and Rainfall (R) to recommend the most suitable crops for a specific region.

By leveraging advanced machine learning (ML) techniques, the system provides highly accurate recommendations, enabling better decision-making. This innovative approach helps farmers increase profitability, and support long-term agricultural sustainability. The proposed model for crop recommendation using Random Forest resulted in 99.78% accuracy, outperforming existing models in predicting suitable crop.

Keywords: Crop Recommendation, Machine Learning, Soil Analysis, Climate Factors

1 Introduction

As technology continues to evolve, the successful sharing of information empowers agriculturists to identify and cultivate their potential. Currently, crop recommendation remains a significant challenge worldwide. A technology-driven crop recommendation system for global agriculture aids farmers in boosting crop yields by recommending the most suitable crops for their land, by considering many factors including geographic and climatic parameters. The suggested hybrid recommendation model has proven to be efficient in identifying an appropriate field. Crop selection is crucial in agriculture, enabling farmers to determine the best crop based on their agricultural land and thereby considering climatic conditions.

ML techniques can greatly assist in automating crop recommendations, helping farmers to enhance their yields while also maintaining soil quality and restoring essential nutrients. Various ML algorithms are utilized for the evaluation of the proposed work. It utilizes various features to accurately identify the most suitable crops for specific regions. This system can transform crop recommendation, helping farmers of all scales by improving crop yields, promoting sustainability, and increasing profitability. By thoroughly analysing a vast dataset, we have attained high accuracy by training and testing models with different parameters.

2 Related works

The authors [1] proposed crop yield prediction using ML models, datasets, and analytical techniques for predicting yield by covering the raw data like climate, soil conditions. By using ML algorithms like Decision Trees (DT), Random Forests (RF), Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Deep Learning (DL) models in predicting crop yields. Examined preprocessing, feature selection, and model evaluation approaches to identify key challenges. Review focused on increasing role of technology in agriculture, stressing the need for high quality datasets and validation techniques to improve accuracy and practical implementation in real-life.

The authors [2] proposed the effects of atmospheric humidity on the growth of horticultural crops. The researchers examined the effects of atmospheric humidity on horticultural crops. By using ML models like SVM, RF and DL models analysed large agricultural dataset, collecting soil factors, climatic conditions, and humidity. These models enhance yield and resource efficiency by identifying suitable crops based

on historical data. Performance is evaluated using various classification metrics and artificial intelligence integrating with real-time make predictions more reliable.

The authors [3] proposed a model for crop yield prediction in the past or the future by using ML models and predicted crop with highest accuracy which is provided by RF, with a Root Mean Square Error(RMSE) of 35-38%,and artificial Neural Networks RMSE of 37-41%,and regularized linear models RMSE of 64-65%.Study used datasets from biophysical crop models, simulating sunflower and wheat yields over 20 years (2001-2020) across five regions in spain, considering soil depth and management practices. The researchers gave importance of proper data splitting for training and validation. Instead, they recommended using validated crop models specific to regions and cultivars before data collection to enhance prediction accuracy.

The authors [4] proposed a model smart Framing by using ML algorithms like logistic regression (LR), DT, RF, KNN, NB, SVM, neural network with highest accuracy 99.5% by model across all models. By considering 100 n-estimators in RF and NB, they predicted crop according to their climatic conditions and soil factors.

The authors [5] proposed a model by utilizing ML algorithms including DT, KNN, Linear Regression, NB, ANN, SVM. In this work, various features had been taken into consideration and achieved accuracy of 89.88% using ANN. This proposed model worked on crop yield for specific factors in climate and soil, in a field by planting single crop or planting multiple crops at a time and also recommends the farmers that the soil is capable to plant a single crop or multiple crops at a time in field by using sensor method.

The authors [6] proposed agroconsultant, an intelligent system, which was deployed to assist farmers about which crops to cultivate beforehand. Agroconsultant recommends suitable crops for cultivation, aiming to enhance crop yield and support farmers in optimizing their agricultural practices. They used various ML algorithms such as SVM, adaboost, bagged tree, ANN, and NB to forecast crop yield.

The authors [7] developed a viable and user-friendly system for crop yield prediction. The approach aimed to mitigate challenges faced by farmers in crop selection, thereby maximizing yield and contributing to the nation's food supply. The work assisted farmers to select or plant a suitable crop based on region's climatic conditions using ML algorithms. Error rate and accuracy were used to assess the model's efficiency. To estimate the amount of rainfall received by a specific location, GPS based location identifier was used.

To improve the crop productivity, the authors [8] recommended a model based on ensemble technique and presented a ML based approach to recommend suitable crops for cultivation. The authors employed an ensemble model that combines predictions from multiple ML algorithms to enhance the accuracy of crop recommendations.

To predict a crop, the authors [9] proposed system for precision agriculture which focuses on developing a system that assists farmers in selecting the most based on various parameters, thereby enhancing agricultural productivity. The authors utilized ML algorithms to analyze factors such as soil composition, climate conditions, and crop requirements. By processing this data, the system can recommend crops that are best suited for specific conditions, aiming to optimize yield and resource utilization.

The authors [10] reviewed in agriculture and provided a comprehensive examination of machine learning applications in agriculture. The model emphasizes transformative role in advancing precision agriculture, particularly in crop recommendation systems.

The authors [11] proposed a model based on machine learning to recommend a suitable crop for farmers to increase the yield of production.The study collects real-time images via an Android mobile application, which are labeled and classified using pre-trained Convolutional Neural Networks (CNNs) such as ResNet18, GoogLeNet, InceptionV3, and MobileNetV3 Large. Among these, MobileNetV3 Large achieved the highest accuracy (93.9%), specificity (96.12%), and precision (97.48%). The study underscores the potential of deep learning for automated plant health monitoring, emphasizing its role in improving agricultural decision-making, optimizing yields, and reducing manual intervention in crop assessment.

To focus on rice crop disease,the authors[12] developed a recommendation system for agricultural products using sentiment analysis. The study employs Natural Language Processing (NLP) and machine learning techniques to assess customer reviews and ratings from agricultural websites. K-Means clustering is applied to categorize products based on sentiment and rating polarity, creating a labeled dataset. The system‘s performance is evaluated using a Support Vector Machine (SVM) classifier, demonstrating moderate accuracy, which validates the clustering approach. This research highlights the importance of integrating sentiment analysis into recommendation systems to assist farmers in selecting optimal disease-control products, ultimately improving agricultural efficiency and productivity.

The authors [13]introduced a machine learning-based crop classification and prediction framework that leverages soil nutrient data to enhance agricultural decision-making. The study assesses multiple ML algorithms, including DT, SVM, NB, KNN, Extreme GB, and RF, to determine the most effective model for crop suitability analysis. Results indicate that Extreme GB and NB perform best, achieving AUC scores of 0.994 and 0.993, respectively. This work underscores the significance of soil health assessment in predictive farming and demonstrates the efficacy of ML algorithms in optimizing crop selection strategies.

To enhance crop the authors [14] proposed an advanced crop recommendation system incorporating an ANN optimized with the Ocean Water Current Optimizer (OWCO). By utilizing Landsat 8 satellite imagery, the study evaluates key agricultural parameters, including soil characteristics, climate variations, and phenological features, to enhance crop selection accuracy. The optimized ANN model achieves a notable accuracy of 94.12%, outperforming traditional metaheuristic models while reducing training time. The research highlights the growing importance of AI, machine learning, deep learning, and bio-inspired optimization in precision agriculture, advocating for computational advancements to refine crop recommendation methodologies.

The authors[15] developed a data-driven smart agricultural system focused on optimizing fertilizer recommendations through big data analytics. The research introduces an Electronic Agricultural Record (EAR) designed to unify disparate agricultural

datasets, enabling precise nutrient management and yield optimization. A large-scale data warehouse, implemented using Hive and Elasticsearch, facilitates real-time analysis and intelligent fertilizer application strategies. This study emphasizes the transformative role of big data in modern farming, advocating for enhanced data integration, real-time decision-making, and resource-efficient agricultural practices.

Gunaganti Sravanthi [16]proposed an advanced IoT-based framework for crop recommendation and disease prediction, aimed at enhancing precision agriculture. The study introduces the Multi-level Kronecker Guided Pelican Convolutional Neural Network (MKGPCNN) for determining the most suitable crops based on key environmental and soil attributes. Additionally, the Combined Graph Sample and Aggregate Attention Network (CGSAAN) is utilized for early disease detection and targeted fertilizer recommendations. The evaluation, conducted on publicly available datasets, demonstrated high accuracy rates of 99% for crop recommendation and 98% for disease prediction, outperforming conventional models. This research highlights the significance of IoT integration with machine learning in modern agriculture, providing data-driven insights to optimize crop selection, improve disease management, and enhance overall productivity while reducing manual intervention.

The authors [17] introduced a machine learning-driven Smart Crop Recommendation (SCR) framework to enhance data-driven decision-making in agriculture. The proposed framework operates in two distinct phases: crop filtration and yield estimation. In the initial phase, an ANN is utilized to assess environmental and soil parameters to filter out the most suitable crops for a given region. The second phase employs a RF regression model to estimate crop yield, considering essential variables such as season, farm area, and geographic conditions. Experimental validation demonstrated high accuracy, with ANN achieving 99.10% in crop selection and RF attaining an score of 0.99 in yield prediction. This study highlights the transformative potential of AI and ML in precision agriculture, emphasizing real-time crop recommendations that optimize productivity while mitigating farming risks.

To strengthen user confidence and enable more informed decision-making in precision farming the authors[18]importance of integrating explainability into AI-driven agriculture.Research underscores the significance of explainable AI (XAI) in advancing crop recommendation systems by enhancing transparency and decision-making. Methods like SHAP, LIME, and ELI5 improve the interpretability of AI predictions.AgroXAI incorporates IoT, edge computing, and ML to generate crop recommendations with clear explanations, promoting user trust. Other studies examine the role of XAI in smart farming, illustrating how improved interpretability facilitates AI adoption. Additionally, ML models integrated with SHAP and LIME enhance reliability by providing well-justified recommendations, bridging the gap between complex AI algorithms and practical agricultural applications.

The authors [19] used Explainable Artificial Intelligence (XAI) in improving transparency and trust in agricultural decision-making. Techniques like SHAP and LIME have been applied to provide visual explanations for AI model predictions, making them more interpretable for users, including farmers and agronomists. Research has demonstrated the effectiveness of these methods in smart agriculture by enhancing the accessibility of AI-driven insights. Additionally, studies have integrated SHAP with

ML models, such as RF to refine crop recommendations and ensure clear justifications for predictions. These approaches not only improve the reliability of AI systems but also foster user confidence in adopting AI-driven solutions for precision farming.

This research explores ML to improve crop recommendation and forecasting. The authors [20] developed a model for maharashtra using historical data, optimizing accuracy with an expectation-maximization technique. The RF algorithm achieved 92% accuracy, while an LSTM network predicted weather three months in advance. Other studies integrate ANN with sensor-based soil data to enhance crop selection and yield. These advancements highlight the potential of ML and DL in creating reliable, data-driven agricultural decision-support systems for sustainable farming.

3 Proposed Model for Crop Recommendation

The "Crop Recommendation System" aims to aid farmers in selecting the most suitable crops for cultivation based on specific climatic conditions like H, T, R and soil properties i.e., contents of N, K, P, pH level of the soil. The system utilizes ML models, particularly supervised learning classification models, to analyse key environmental elements to provide tailored crop recommendations.

Objective of the model is to enhance agricultural productivity guiding farmers to make data-driven decisions regarding crop selection. Using supervised learning-based ML models, the system is trained on data set provided which has environmental elements and crop yields. The model predicts the most suitable crops for given conditions, ensuring better resource utilization and optimal yields.

The proposed crop model primarily involves use of four stages as shown in Fig 1. 3.1: The "Crop Recommendation" dataset used for the proposed model is taken from Kaggle, during the data collection phase. Crop recommendation dataset consists of various features like soil properties, climatic conditions and target labels such various crops. 3.2: Data preprocessing and selecting ML model. 3.3: Training and making predictions based on testing data. 3.4: Performance evaluation.

3.1 Data Collection

Critical features from the dataset are extracted and included, which are essential for predicting the most suitable crops for given environmental and soil conditions. The primary objective is to utilize these features for supervised learning tasks, enabling accurate crop recommendations. This dataset is organized in a data frame format, commonly used in python's panda's library for efficient data analysis and manipulation. The structure likely includes various columns that represent environmental conditions, soil properties, and target labels for crop classification. Various crop types serve as the target output, helping the farmers to make informed crops for specific conditions. The description of the dataset is given in Table 1.

3.2 Data Preprocessing

Data preprocessing is an important phase and plays a vital role in achieving outstanding results by ensuring that the dataset is noise free and ready for analysis. In this

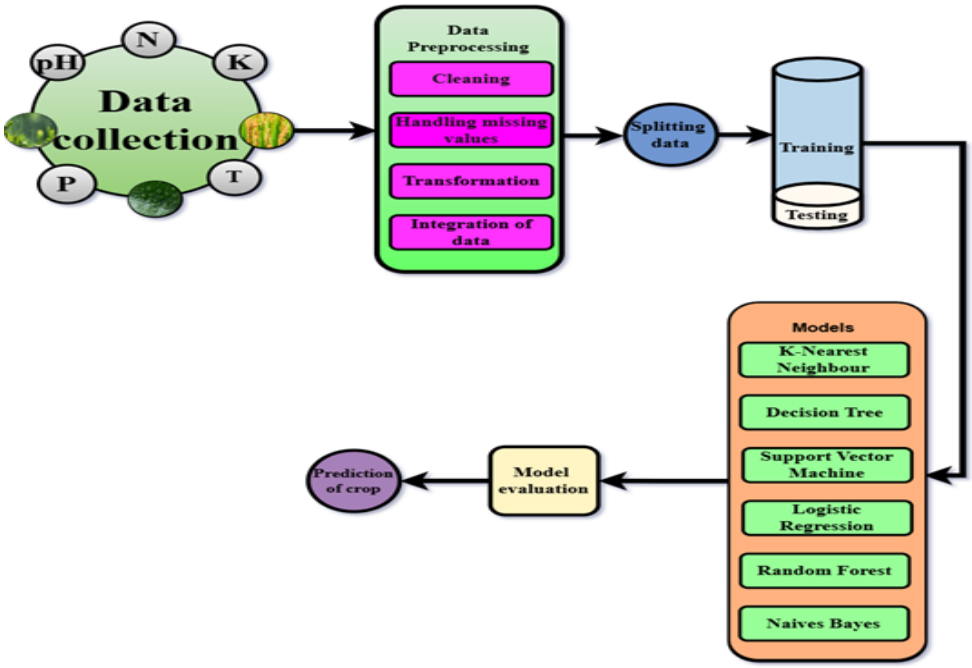

Fig. 1: **Workflow of The Proposed Crop Recommendation Model**

Table 1: Samples from the dataset

S.no	N	P	K	T	H	pH	R	Label
1	40	78	20	19.1856	20.8339	5.6693	80.152934	Kidney beans
2	32	57	22	28.689985	87.504368	6.769416	44.565984	Mung beans
3	20	77	23	34.872487	38.837860	5.180272	148.250279	Pigeon peas
4	14	8	43	21.925139	94.464853	7.051655	111.716202	Pomegranate
5	29	11	5	23.133388	91.946703	7.639788	104.422415	Orange
6	90	52	25	25.974824	69.363857	6.822587	103.223421	Maize
7	89	83	47	28.095776	77.795868	5.631272	109.540861	Banana
8	10	125	196	22.312537	90.035771	5.730557	113.068815	Apple
9	111	5	47	28.033065	91.473558	6.274453	21.179248	Muskmelon
10	115	11	46	24.415927	89.396555	6.623167	40.321619	Watermelon

work, the target labels are strings. To facilitate efficient data analysis, target labels are converted from strings to numerical values (encoded values) process called label encoding, and the label encoding Table 2 is attached here for reference.

3.3 Training and Testing of the Proposed Model

The bifurcation of the dataset resulted in train and test sets with 80% and 20% to train and evaluate the performance. After training, the model is evaluated using the test

Table 2: Samples from the Dataset

Crop	Encoded Values
Rice	1
Maize	2
Chick pea	3
Apple	4
Orange	5
Papaya	6
Coconut	7
Jute	8
Cotton	9
Kidney beans	10
Pigeon peas	11
Coffee	12
Watermelon	13
Pomegranate	14
Muskmelon	15
Mung bean	16
Moth beans	17
Mango	18
Lentil	19
Grapes	20
Black gram	21
Banana	22

data, which is separated from the original dataset. Supervised learning model includes the independent variable 'X' and dependent variable 'Y' and the mapping is done from 'X' to 'Y' by analyzing the relationship between these variables. The model tries to learn the complex relationship among the variables, which helps in proper predictions.

$$Y = f(X) \tag{1}$$

The model is trained to minimize the error by approximating the map function given in Eq. 1. For the crop recommendation dataset, multiple ML algorithms are utilized to classify the data and to predict the suitable crop.

3.3.1 K-Nearest Neighbor

KNN algorithm predicts the label based on the nearest neighbors to the new instance using distance metrics like Euclidean distance. In classification, the label of a new data point is assigned based on the most common class among its 'K' nearest neighbors. It is simple, nonparametric, and effective, but not well suitable for large datasets as the model initially have to calculate the distances from each data point. Below table shows the accuracy achieved using KNN model with different values. Euclidean distance is effective in KNN as it measures the direct shortest distance between points. It is widely used in ML when features are normalized due to its efficiency and reliability.

The formula for Euclidean distance metric is given by,

$$d\left(j,k\right)=\sqrt{\sum_{i=1}^{n}\left(j_i-k_i\right)^2} \tag{2}$$

where j, k are the components of x, y coordinates, j_i, k_i are the ith components of x, y coordinates.

Table 3: Accuracy Of the KNN Model with Different 'K' Values (Euclidean Distance)

S.no	K Value	Accuracy
1	3	97.88%
2	5	97.58%
3	7	97.27%
4	9	97.42%

As shown in Table 3, the KNN algorithm achieves its highest accuracy of 97.88% at K=3.

3.3.2 Support Vector Machine

Both classification and regression problems can be solved using SVM algorithm. It creates a hyperplane separating the data into two categories by extending the margin among them. The margin is defined as the distance between the hyperplane and the nearest data points from each class, known as support vectors. Kernel functions are used, which makes it suitable algorithm for both linear and non-linear relationships. Hyperplane equation in non-linear classification,

$$\omega\cdot\phi\left(x\right)+b=0 \tag{3}$$

where x represents the input feature vector, $\phi\left(x\right)$ defines mapping function, w denotes the weight vector, and b is the bias term.

Since data in the dataset is of linearly not separable, kernel PCA is used for the dimensionality reduction. To map the data points into high dimensional space, a technique called "kernel PCA" is used, enabling the capture of non-linear relationships. It finds the principal components in this transformed space, revealing complex patterns that traditional methods may not uncover. This approach is ideal for non-linear dimensionality reduction.

3.3.3 Logistic Regression

Logistic regression is especially used for classification problems. This algorithm makes use of sigmoid function to estimate probabilities:

$$\sigma = \frac{1}{1 + e^{-w}} \tag{4}$$

$$w = \alpha \cdot i + b \tag{5}$$

where, w denotes the weighted sum, α is coefficient, i represents feature vector, and b is the bias term. In binary classification, a threshold (e.g., 0.5) determines class assignment.

Since dataset consists of continuous features and 22 class labels, we employ multi-class logistic regression, extending the binary case via:

- **One-vs-Rest (OvR)** : Trains 22 separate binary classifiers, each distinguishing one class from the rest.

3.3.4 Decision Tree

DT works by recursively splitting the dataset based on feature values to create a tree, consisting of internal nodes, branches, and leaf nodes representing condition/decision possible outcomes, and class labels respectively. Here in the proposed model, to make decisions a criterion called "information gain" is used to make a particular feature as root node. From this node, the tree evolves to the leaf nodes. Evaluated sample decision tree with some selected crops can be seen in Fig. 2

- **Splitting Criteria (Handling Continuous Features)**:
  Since dataset has continuous features, the decision tree must find the best split point for each feature. It does this by evaluating multiple thresholds for each feature and selecting the one that best separates the data. Common split criteria:
- **Entropy**: It is a measure of randomness or impurity in the dataset:

$$H(X) = -\sum_{i=0}^{n} Q_i \log_2 (Q_i) \tag{6}$$

  where X defines variable taken from dataset, Q_i is the probability of occurrence of the ith outcome, n defines number of possible outcomes.
- **Weighted Entropy After Split**:
  After splitting on attribute, A, entropy is calculated for each subset

$$H\left(\frac{X}{Y}\right) = \sum_{j=1}^{m} \frac{|X_j|}{X} H(X_j) \tag{7}$$

  where m represents number of subsets.
- **Information gain**:
  It quantifies how much entropy decreases when we split a dataset using particular feature. It's a measure used in decision trees to decision which feature to split on.

$$IG(X, Y) = H(X) - \sum_{j=1}^{m} \frac{|X_j|}{|X|} H(X_j) \tag{8}$$

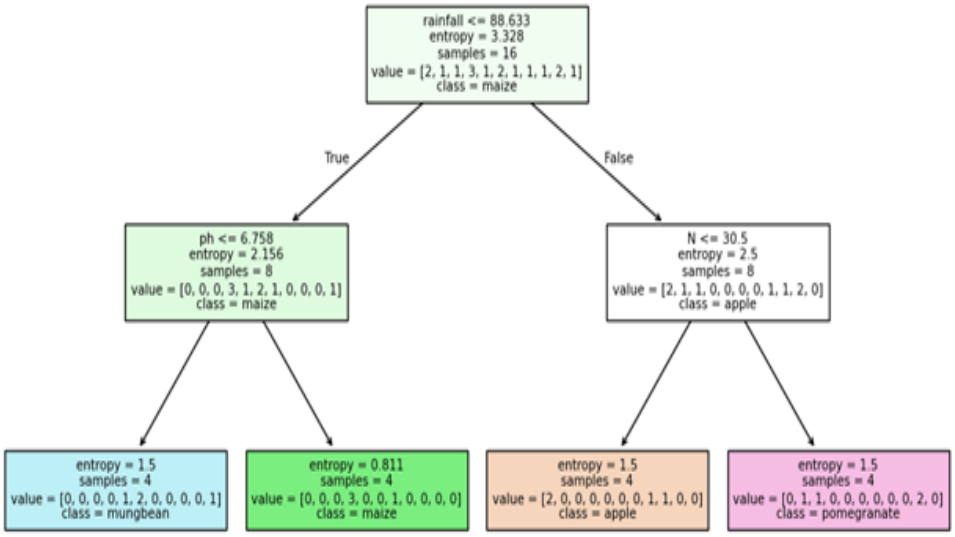

Fig. 2: Sample Decision Tree for Selected Crop Classes

3.3.5 Random Forest

RF is a popular ensemble technique that builds multiple DTs during the training process and combines the trees' outputs to improve the accuracy and reduce overfitting.

In the RF model, dataset, consisting of 22 discrete labels and 7 continuous features, is split using the bootstrap aggregation (bagging) technique. A random subset of the dataset is chosen and assigned to each tree for training, typically containing around 67% of the samples, selected with replacement. The remaining 33% of the data (Out-of-Bag samples) are left out and can be used for model validation. Additionally, at each node split within a tree, only a random subset of features is considered to reduce correlation between trees. The best split is determined using entropy (information gain), which helps in selecting the most informative feature and threshold to separate the classes effectively. This process ensures that each tree captures different patterns, reducing overfitting and improving the model's generalization.

- **Mathematical Representation of Random Forest Classifier**
  Given a dataset D with 2200 samples and 7 features, Random Forest:
  Creates T decision trees $f_1, f_2, f_3, f_4, \ldots f_t$.
  A random subset is assigned to each tree for training from the entire set.

The predicted value for the new input x is

$$Y = \text{mode}\left\{f_1\left(x\right), f_2\left(x\right), \ldots, f_t\left(x\right)\right\} \tag{9}$$

3.3.6 Naive Bayes

The NB algorithm is based on bayes' theorem, which is used to solve classification problems, assuming all features are independent of each other, simplifying computation. The algorithm calculates the probability(pro) of a sample belonging to a class by combining the probabilities of its features. It is fast, efficient, and suitable for large datasets, but its accuracy can be affected when feature dependencies exist. As our features are continuous, by gaussian naive bayes the probability estimation is:

$$Pro\left(\frac{A_j}{Z}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(a_j-\mu)^2}{2\sigma^2}} \tag{10}$$

where, A_j represents the feature in the dataset, a_j represent specific datapoint, σ is the varience, μ is mean value. For classification, the probability(Pro) of features B belonging to a class Z_m is calculated by using bayes theorem as:

$$Pro\left(\frac{Z_m}{B}\right) = \frac{Pro\left(\frac{B}{Z_m}\right) P\left(Z_m\right)}{Pro\left(B\right)} \tag{11}$$

$$Pro\left(\frac{Z_m}{B}\right) = Pro\left(\frac{N}{Z_m}\right) Pro\left(\frac{Pro}{Z_m}\right) Pro\left(\frac{K}{Z_m}\right) Pro\left(\frac{T}{Z_m}\right) Pro\left(\frac{H}{Z_m}\right) Pro\left(\frac{R}{Z_m}\right) Pro\left(\frac{pH}{Z_m}\right) \tag{12}$$

The probability of selecting a crop is:

$$Pro\left(\frac{B}{Z_m}\right) \propto Pro\left(Z_m\right) \prod_{j=1}^{n} Pro\left(\frac{b_j}{Z_m}\right) \tag{13}$$

3.4 Performance Evaluation

Confusion matrix is used to assess the performance of the models used in the proposed work. Fig 3 shows the confusion matrix generated for the RF classifier in the proposed work, and the performance of the various models can be evaluated using a confusion matrix, which consists of True positive ($TP_{(corr)}$), False positive ($FP_{(err)}$), True negative ($TN_{(corr)}$), False negative ($FN_{(err)}$) and evaluates the model's performance by comparing the predicted classifications with the actual class labels found in the ground truth.

Performance of a model can be evaluated by using different metrics in classifications. These metrics give insight into how well the model predicts cases by assessing various aspects of its predictions. Each metric highlights a different perspective of the model's performance, helping to identify strengths and weaknesses.

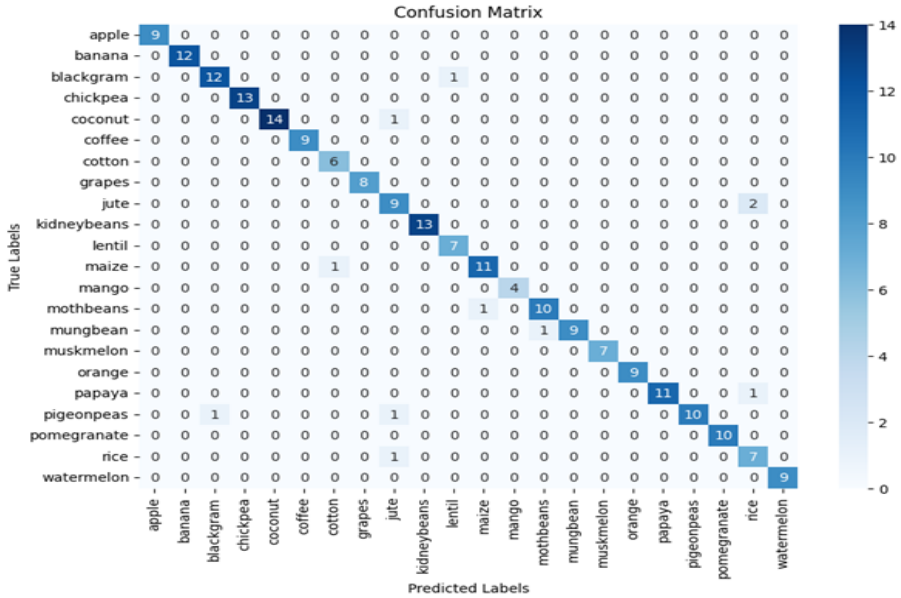

Fig. 3: Confusion Matrix of Random Forest classifier for the Multiclass Classification

3.4.1 Accuracy

Accuracy (Acc) is one of the performance-evaluation metrics while evaluating predictions. Percentage of correct predictions by a model in comparison to total predictions. Accuracy achieved by using different ML models shown in Fig 4. Accuracy is a valuable and easily interpretable metric, increase in accuracy also denotes that the model is classifying the right pieces much better for all classes.

$$\text{Accuracy} = \frac{TP_{(corr)} + TN_{(corr)}}{TP_{(corr)} + TN_{(corr)} + FP_{(err)} + FN_{(err)}} \tag{14}$$

3.4.2 Precision

Precision (Pr) quantifies number of positively predictive results are actually positive. A higher precision indicates fewer false positives and a more reliable positive prediction and the precision generated by using various ML models is shown in Fig 5.

$$\text{Pr} = \frac{TP_{(corr)}}{TP_{(corr)} + FP_{(err)}} \tag{15}$$

3.4.3 Recall

Recall (Re) is the proportion of actual positive cases that the model successfully predicts as positive. Recall attained by using different ML models is shown in Fig 6.

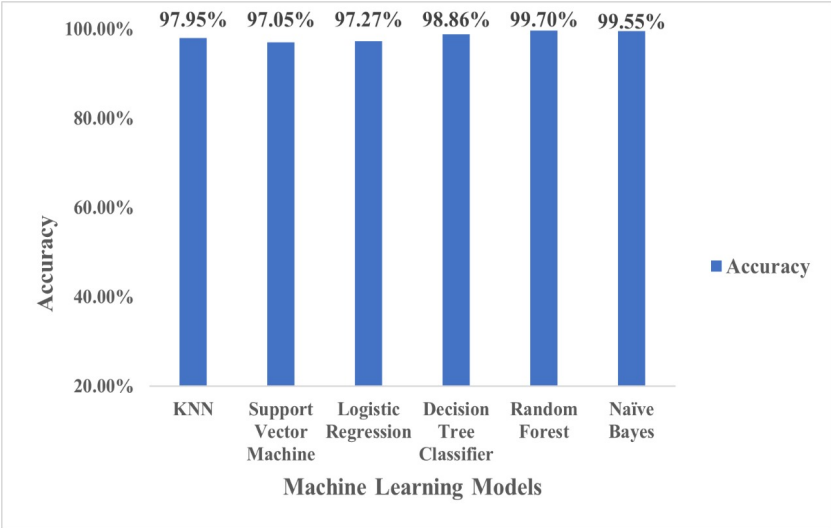

Fig. 4: Accuracy for different models used in the proposed work

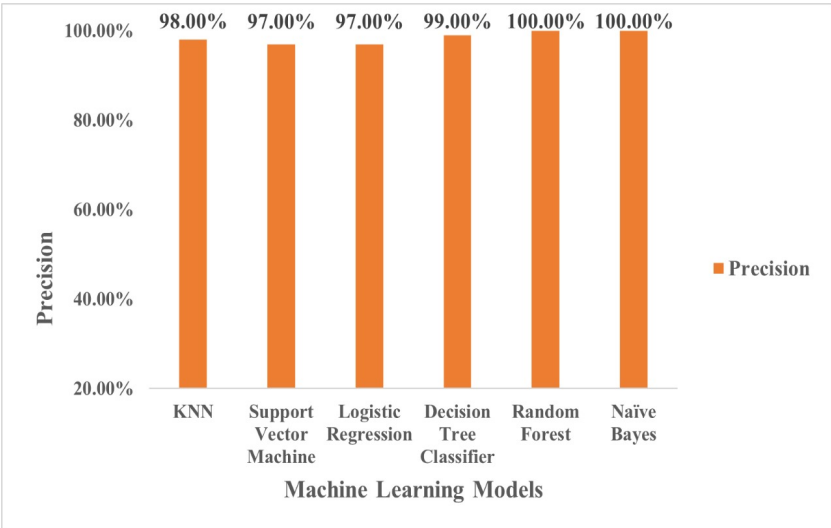

Fig. 5: Precision for different models used in the proposed work

Mathematically, recall is calculated as:

$$\text{Re} = \frac{TP_{(corr)}}{TP_{(corr)} + FN_{(err)}} \tag{16}$$

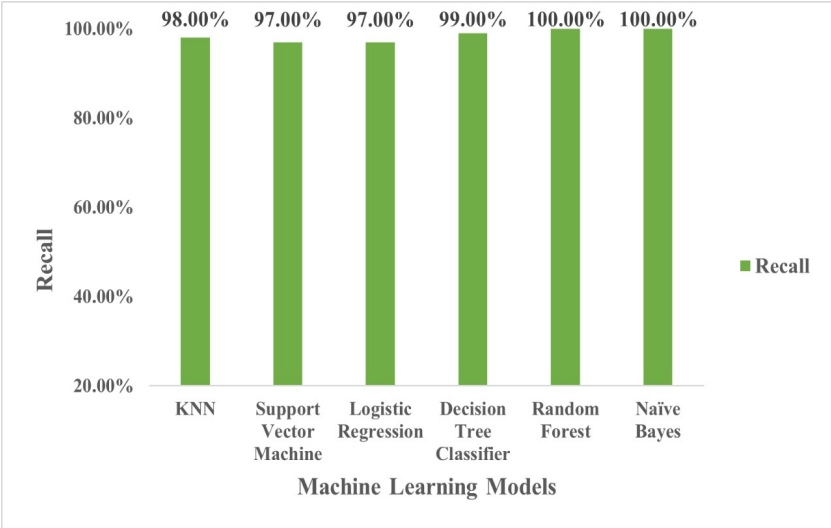

Fig. 6: Recall for different models used in the proposed work

3.4.4 F1-Score

The F1-Score is the harmonic mean of precision and recall. The score ranges from [0, 1]. Fig 7 shows the F1-Score attained for various machine learning models with our dataset. The formula for the F1- Score is:

$$F1_Score = \frac{2 \times \mathrm{Pr} \times \mathrm{Re}}{\mathrm{Pr} + \mathrm{Re}} \tag{17}$$

3.4.5 AUC ROC Curve

Area under the curve -receiver operating characteristic curve is used for evaluating the performance of binary classification models and plots the true positive rate (TPR) against false positive rate (FPR) at various thresholds showing difference between two classes such as positive and negative outcomes. TPR is also called as sensitivity or recall, FPR also known as specificity. TPR is defined as the ratio of correctly predicted actual positives and FPR is defined as the ratio of incorrectly predicted actual negatives. TPR and FPR terms are derived from confusion matrix, which provides the $TP_{(corr)}$, $TN_{(corr)}$, $FP_{(err)}$, and $FN_{(err)}$ values. AUC measures the area under the ROC curve, high values of AUC give better model performance, value of 1.0 indicates the perfect performance, and 0.5 value of AUC indicates random guessing. The AUC-ROC curves attained for various models used in the proposed works are shown in Figures 8, 9, 10, 11, 12, and 13 for KNN, SVM, LR, DT, RF, and NB respectively.

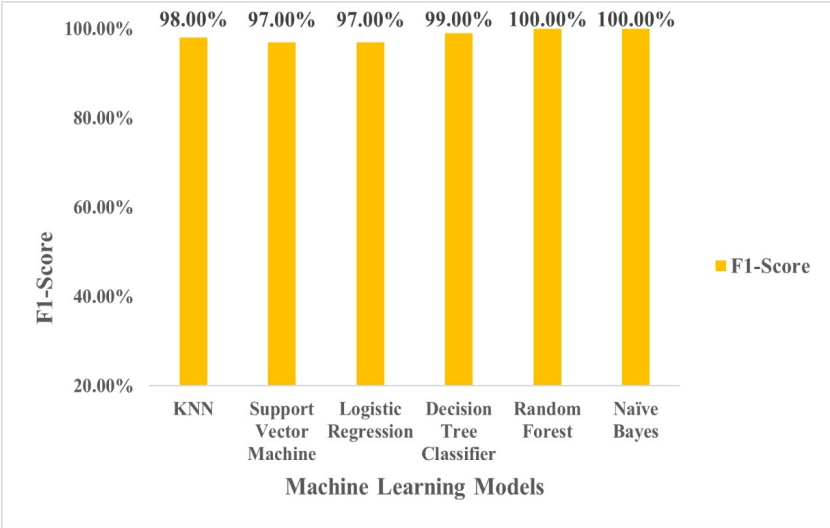

Fig. 7: F1-Score for different models used in the proposed work

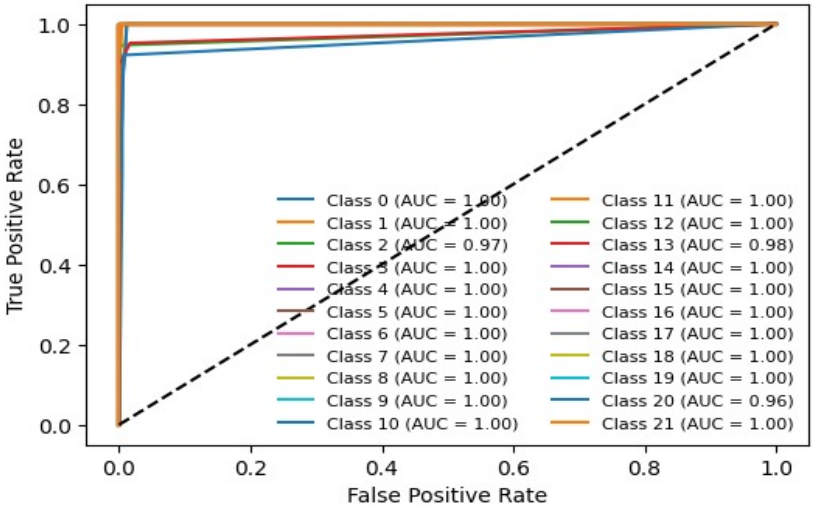

Fig. 8: AUC-ROC Curve of KNN

4 Hyperparameter Tuning Using Bayesian Optimization Technique

There are two types of parameters in the ML model

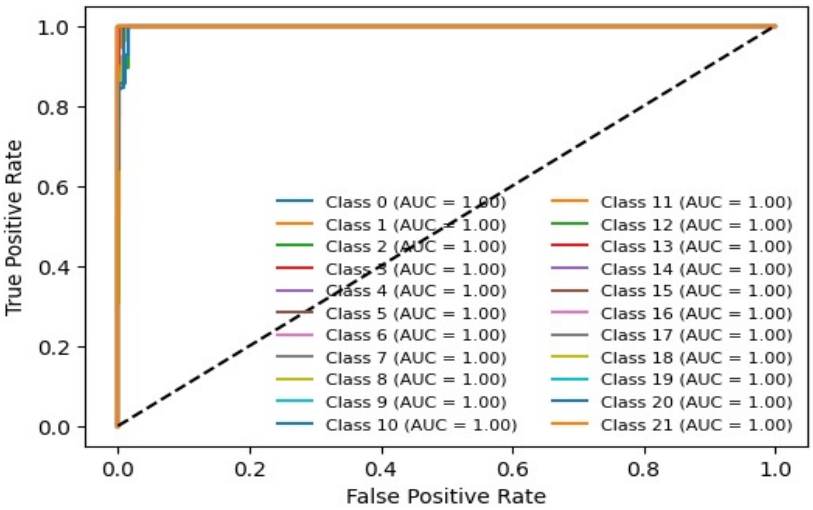

Fig. 9: AUC-ROC Curve of SVM

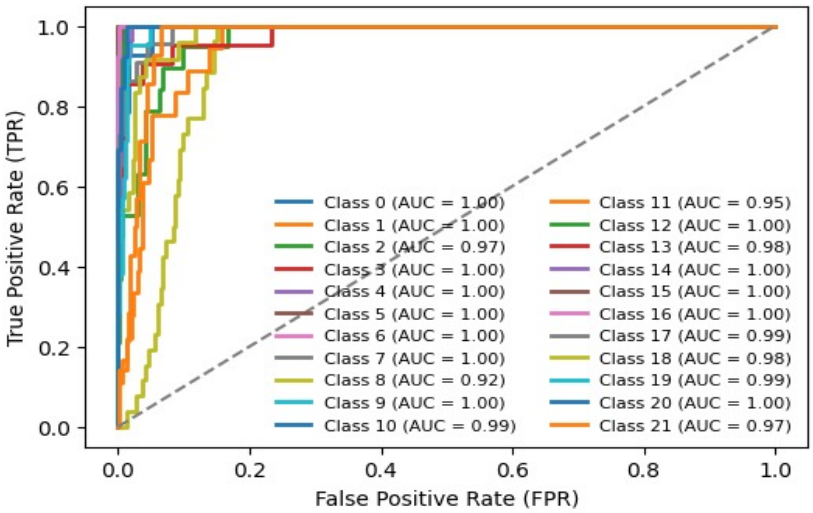

Fig. 10: AUC-ROC Curve of LT

4.1 Model Parameters

Variable of selected model which can estimated by fitting the given data to model. Values are learned from the dataset during training. Model parameters are required for making predictions and are estimated by optimization algorithms like gradient descent, adam, adagrad.

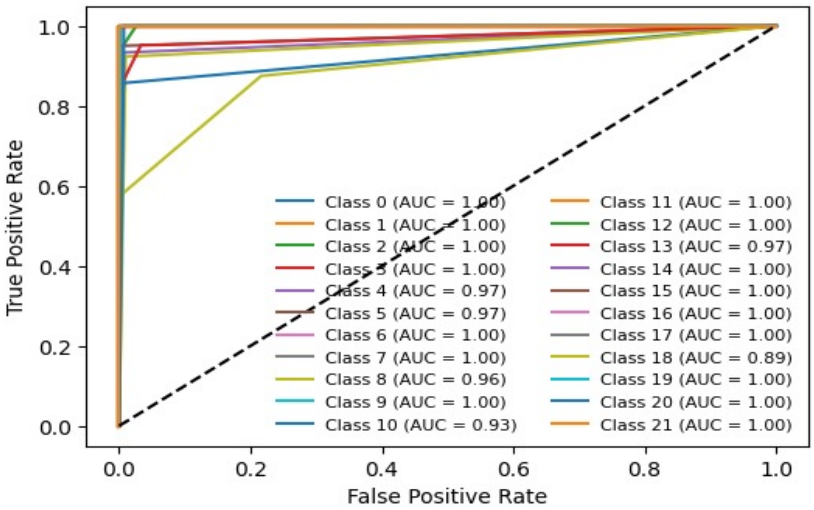

Fig. 11: AUC-ROC Curve of DT

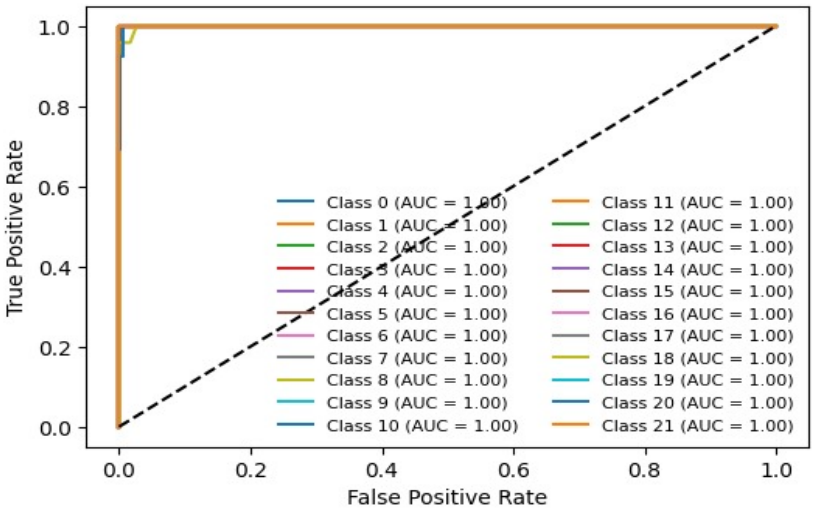

Fig. 12: AUC-ROC Curve of RF

4.2 Hyperparameters

Parameter whose value is set before the model training starts, and they are not learned by fitting the model to data. Hyperparameters are required for estimating the model parameters and estimated by using hyperparameter tuning techniques. Specifically, hyper parameter tuning technique Bayesian optimization is used, as it is particularly effective when dealing with expensive, noisy, and black-box functions and gives high

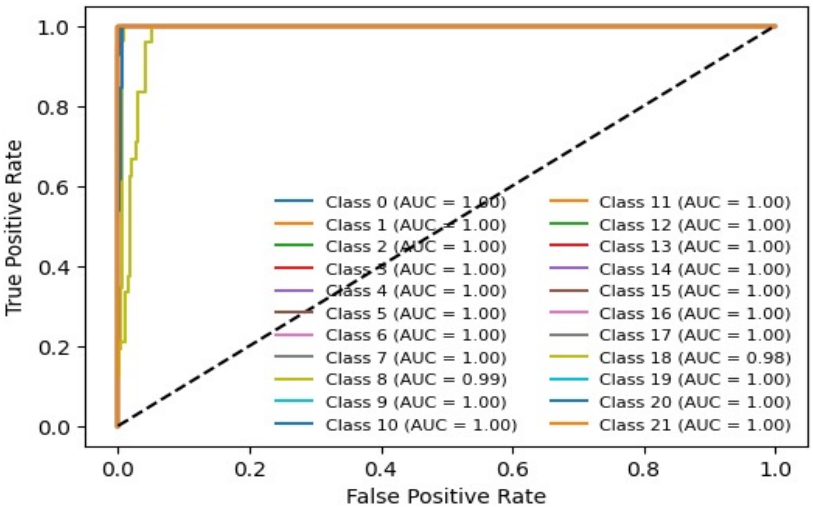

Fig. 13: AUC-ROC Curve of NB

efficiency by considering minimal number of evaluations and can be applied to wide range of optimization problems. This optimization involves two main components like surrogate model and acquisition function. Table 20. shows the optimized accuracy, optimized precision, optimized recall, optimized F1-score after applying bayesian optimization hyper parameter tuning technique on ML algorithms along with best performance achievable configurations.

In KNN the nearest neighbours, weights, and metric are used as hyper parameters to find the optimized accuracy after applying hyper parameter tuning technique bayesian optimization. N-neighbours are defined as nearest neighbours which are checked when an item is being classified. Lower n-neighbours values lead to overfitting and higher values lead to underfit so considering n-neighbours value as eight is best choice weights consists of parameters like uniform which can be defined as all neighbours have equal importance and distance which can be defined as closer neighbours have higher weight and third parameter used is metric which can be defined as the distance between two points and there are three types of distance follow as euclidean distance, manhattan distance, minkowski distance. Manhattan distance is taken as best choice because it works well for all continuous numerical dataset. Fig 14. shows the AUC-ROC curve for KNN.

In SVM kernel, C, and gamma are used as hyperparameters to define the optimized accuracy using bayesian hyper parameter tuning technique. Lower values of C contain more flexible margin and less overfitting, higher values of C give more prone to overfitting. By considering C value as 16.2984 can be considered as best choice as it has the balance between generalization and complexity. Kernel has three components linear, rbf, poly in which linear is considered as best choice as it works well for simple data and it can be separated linearly without complex boundaries. Gamma has two

Table 4: Optimal parameter values for the proposed ML models

Sl. No	Model	Configurations	Optimal Value	Acc	Pr	Re	F1-Score
1	K-NN	N_Neighbours: Range(1,20)	N_Neighbours:8	98.41%	98.59%	98.41%	98.40%
		Weight:[Uniform, Distance]	Weight: Distance				
		Metric:[Euclidean, Manhattan, Minkowski]	Metric: Manhattan				
2	SVM	C:Range(0.01,100)	C:16.2984	98.86%	98.91%	98.86%	98.86%
		Kernel:[Linear,Rbf,Ploy]	Kernel: Linear				
		Gamma:[Scale,Auto]	Gamma: Auto				
3	LR	C:Range (0.01,100)	C:14.410	96.14%	96.44%	96.14%	96.19%
		Penalty:[L1,L2]	Penalty:L1				
		Solver:[Liblinear,Saga]	Solver: Liblinear				
4	DT	Criterion:[Gini,Entropy]	Criterion: Entropy	98.63%	98.48%	98.41%	98.41%
		Min_Split: Range (2,10)	Min_Split:7				
		Max_Depth (10,50)	Max_Depth:24				
		Min_Leaf: Range (1,4)	Min_Leaf:1				
5	RF	N-Estimators: Range(50,300,50)	N_Estimators:214	99.77%	99.78%	99.77%	99.77%
		Max_Depth:Range(1,50)	Max_Depth:17				
		Min_Leaf:Range (1,4)	Min_Leaf:5				
		Min_Split:Range(2,5)	Min_Split:3				
6	NB	Var_Smoothing:(1e-9,1)	Var_Smoothing: 5.853883214e-07	99.55%	99.59%	99.55%	99.54%

parameters scale, and auto which can be defined as how much influence a single training example has and auto is considerd as best choice as it gives good generalisation by using 1/ (number of features) where scale uses 1/ (number of features* variance). Fig 15. shows the AUC-ROC curve for optimized SVM. Hyperparameters used in LR are C, penalty, Solver. C parameter defines as used to control regularization strength, the lower values of C give more regularization and better generalization, higher values of C give less regularization and uses more complex model. Considering C value as 14.56 suggests the increase in accuracy with some regularization. Lasso(L1) and Ridge(L2) are the parameters of Penalty in which L1 is considered as best to improve accuracy by selecting some important features as it shrinks some coefficients to zero and also considered as good for feature selection where as L2 shrinks all coefficients equally by preventing from overfitting. Liblinear solver is considered as it indicates dataset is not too large, and in case L1 penalty is preferred. Fig 16. show the AUC-ROC curve for optimized LR.

Criterion, Min_leaf, Min_split, Max_depth can be considered as hyperparameters in DT classification to achieve the optimized accuracy. Criterion determines how the

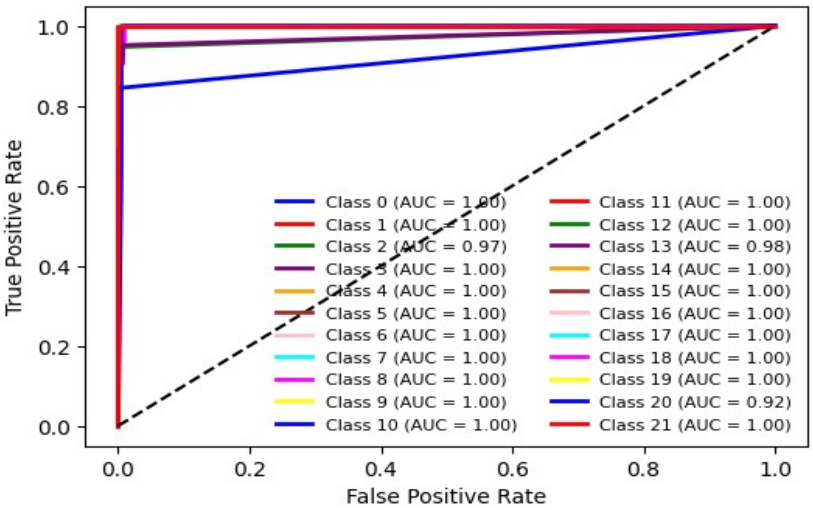

Fig. 14: AUC-ROC Curve of Optimized KNN

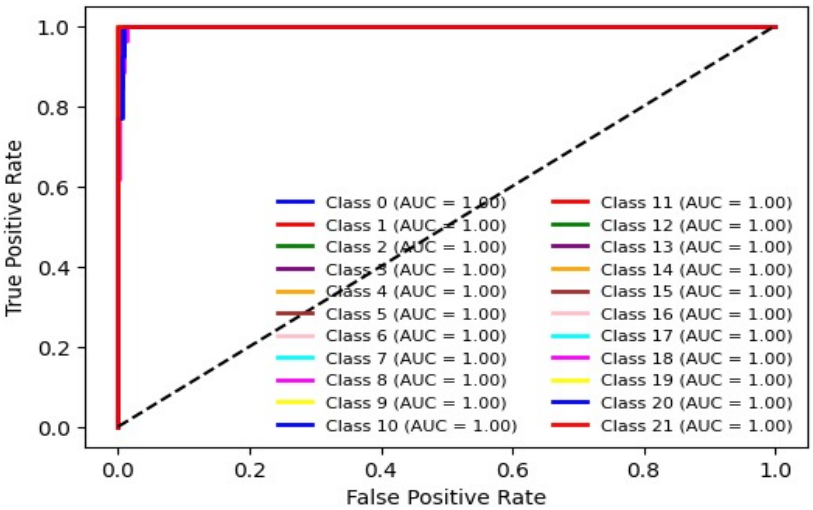

Fig. 15: AUC-ROC Curve of Optimized SVM

model decides the best split at each node in which gini and entropy are models in criterion where entropy is considered, as it is better for datasets which has imbalance classes or more informative splits by using information gain to determine the best split. Minimum samples leaf gives the minimum number of samples a node must have to be a leaf, as smaller values allow deep trees considering Min_leaf value as one is better. Min_split can be defined as number of samples needed to split a node. Considering

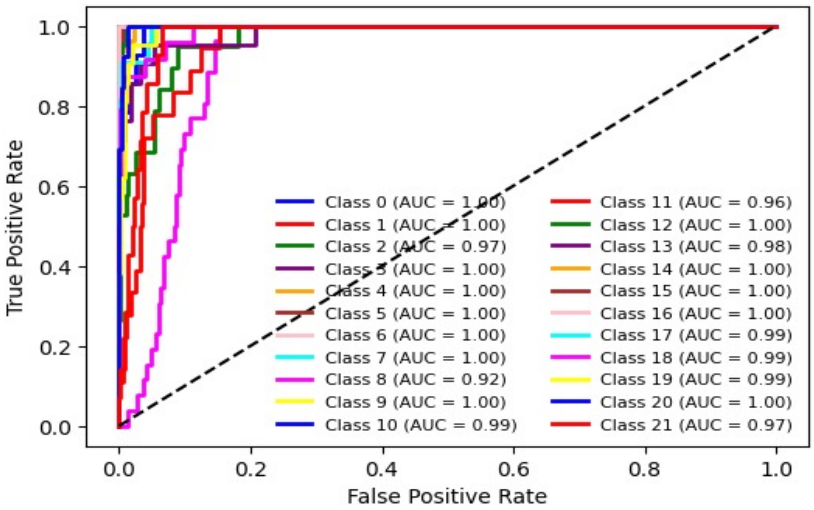

Fig. 16: AUC-ROC Curve of Optimized LR

value as seven is best choice to prevent from unnecessary splits. Max_depth defines as number of splits a tree can make, in which higher value can overfit and lower value may underfit. Max_depth value as two and min_leaf as value one gives best where it controls the complexity while maintaining accuracy. Fig 17. show the AUC-ROC curve attained for optimized DT.

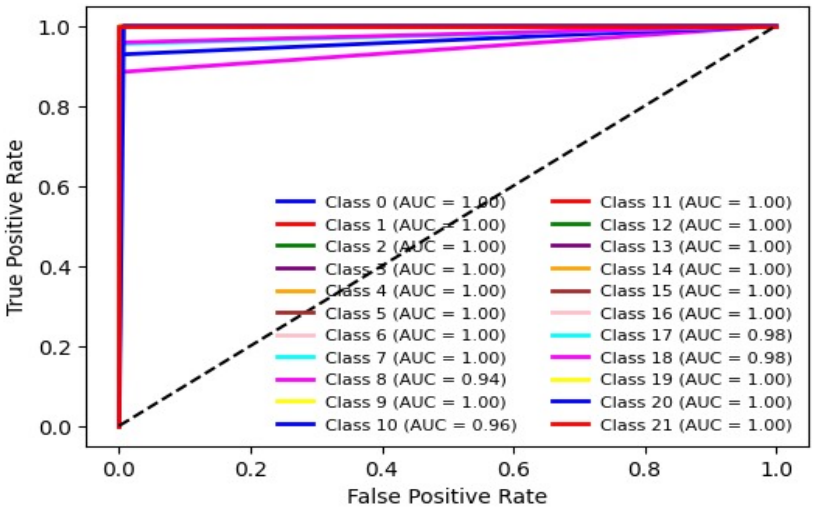

Fig. 17: AUC-ROC Curve of Optimized DT

RF is the combination of more than one decision tree. Combination of many decision trees increases the complexity, training time, generalization, unstable predictions. To get rid of these we use n-estimators, max_depth, min_leaf, min_split. N-estimators defines that how many decision trees are combined to form random forest. Max_depth defines that how many divisions a tree can make from the root to a leaf. Min_leaf defines that how many instances are required for each leaf. Min_split defines that minimum number of samples are required to split the internal node. If n-estimators are too low then the model cannot cover all the patterns and having high value at the beginning increases performance but after certain point improvement becomes too small and increases the calculation time and having two estimators is good for accuracy and efficiency. If the max_depth is too small it may lead to underfit and it may lead to overfit if it is too high and max_depth of six without memorizing the noise it captures main patterns. Fig 18. gives curve for optimized RF.

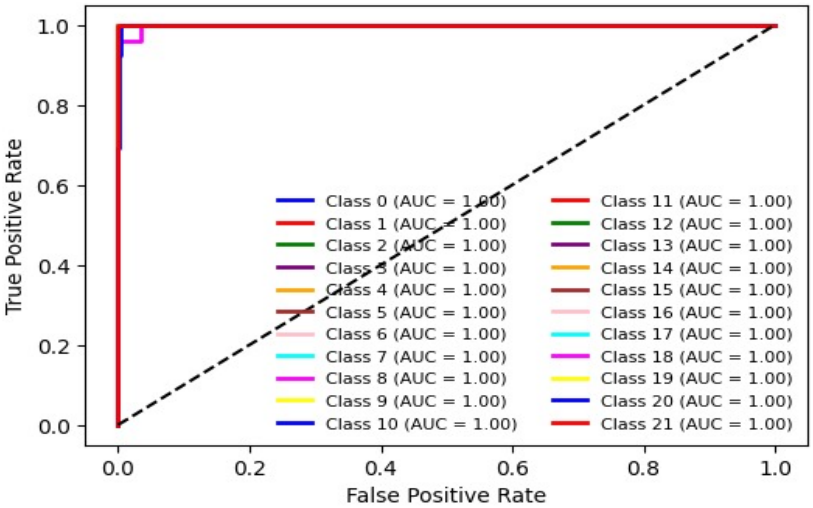

Fig. 18: AUC-ROC Curve of Optimized RF

Var_Smoothing uniform hyper parameter is used in NB model to achieve the optimized accuracy by using bayesian optimization hyper parameter tuning technique.it can be defined as adding a small number to variance estimates to prevent from zero probabilities. NB performs best when var_smoothing uniform value 5.85388321494e-07 is taken in between range of (1e-9,1) is finely adjusted to avoid numerical issues. Fig 19. shows the AUC-ROC curve for optimized NB.

5 Results and Discussions

The confusion matrix provided insight into the classification ability of each model, highlighting the reliability of RF in predicting the most suitable crops. Other models

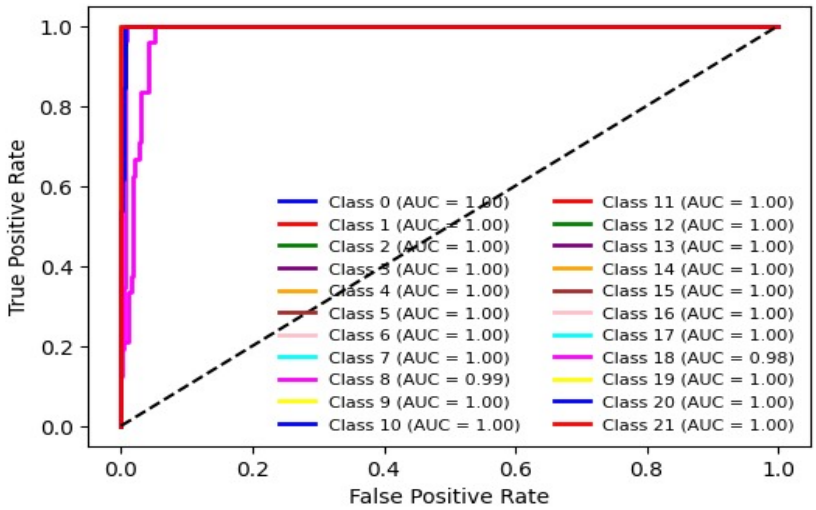

Fig. 19: AUC-ROC Curve of Optimized NB

like DT and NB also delivered competitive performance, but they fell short of the efficiency demonstrated by RF in this specific application. Performance evaluated by each algorithm is shown in Fig 20.

The proposed model identifies the most suitable crops for a given region with an accuracy of 99.78% using RF which surpasses the performance of existing crop recommendation systems.

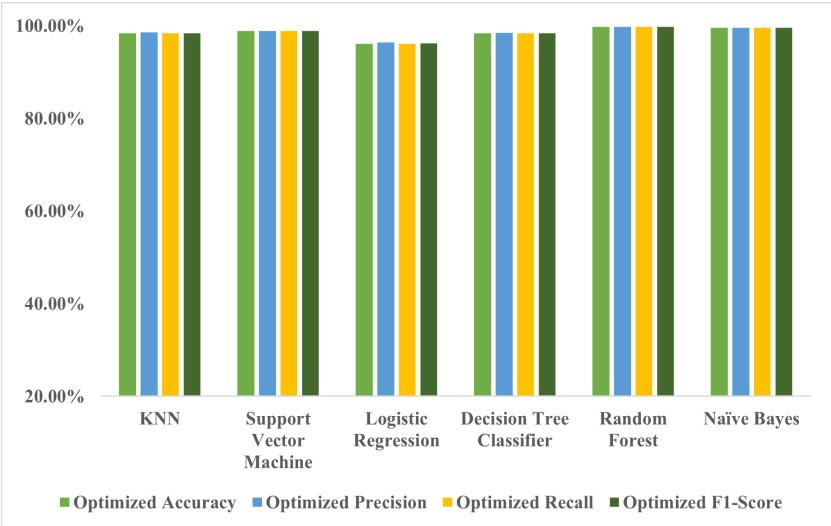

Fig. 20: Performance Analysis of Optimized Classifiers

The ML models utilized in the proposed system were evaluated using various metrics. Among the six models tested, RF exhibited the best accuracy of 99.78% and surpassing all other algorithms in terms of precision and recall.

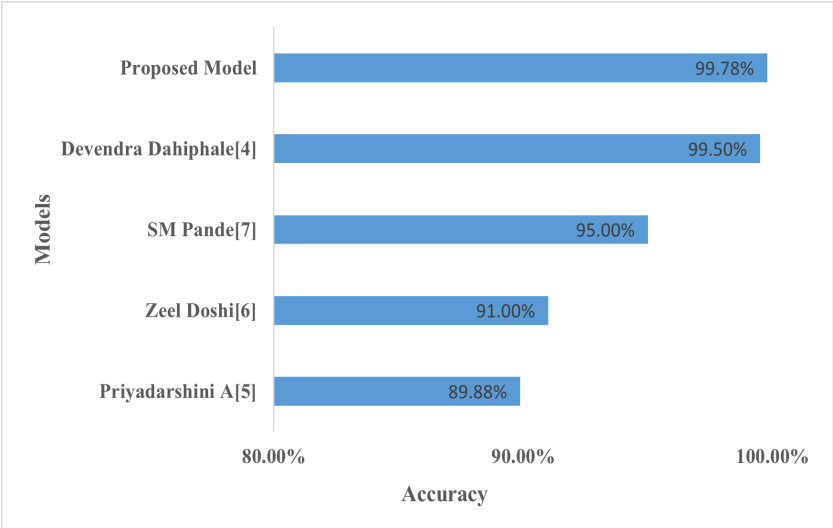

Fig. 21: Accuracy Comparison of Proposed Model with Existing Models

A RF with 214 n-estimators achieved highest accuracy 99.78% by using bayesian optimization hyper tuning technique. Estimators with highest values leads to reduce in variance. As the value of n_estimators increases, the increase in accuracy can be seen compared with low values. In comparison with other works, they used n_estimators with value 100 and attained accuracy is 99.5%. Increase in n_estimators values will not cause to overfitting and gives best accuracy which is more stable and efficient. Crop recommendation system employs maximizing crop yield, sustainable usage of resources, adjusting to changing weather, avoiding crop failures by protecting the environment as it is essential for farmers. The proposed model is to recommend best crop based on several conditions of N, K, P, H, T, pH and R. This study demonstrates how data-driven systems can empower farmers to make informed decisions, ultimately promoting sustainable agricultural practices. Future work may focus on expanding the dataset, incorporating additional features such as pest presence or market trends, and exploring ensemble techniques to further enhance prediction accuracy.

6 Conclusion

The crop recommendation system using ML models has shown to be very good at helping farmers to choose the best crop based on important factors like weather and soil conditions. After testing many different models, the RF model performed the best, with an accuracy of 99.78%, better than other models like DT and NB. The system

used different measures like accuracy, precision, recall, and F1-score to check how well the model worked, and RF always gives the best results for predicting the right crop. This model looks at many factors like N, K, T, and R to help farmers grow the best crops, use resources wisely, and improve their harvest. This means it helps farmers avoid crop failures and protects the environment by being efficient. In the future, the system could get even better by adding more data, such as information about pests or market prices, and trying new methods to make predictions even more accurate. This shows how ML can help farming become smarter and more sustainable. The experiment shows that our crop recommendation model works better than older models from past studies. The results prove that our model gives the best recommendations for choosing the right crop for farmers.

Author's Statement

- Funding: The authors received no financial support for the research and/or publication of this article.
- Conflict of interest/Competing interests: The authors declare that they have no conflicts of interest related to this work.
- Ethics approval and consent to participate: Not Applicable.
- Consent for publication: Not Applicable.
- Data availability: The dataset used in this study is publicly available.
- Materials availability: The study did not use any additional materials requiring availability disclosure.
- Code availability: The code supporting the research findings is available with the corresponding author.
- Author contributions: Prakash Mogga: Conceptualization, Methodology, Writing Original Draft.
  Amrutha Varshini Thanneeru: Conceptualization, Methodology, Writing Original Draft.
  Lopinti Hari: Conceptualization, Methodology, Writing Original Draft.
  Shyam Sunder Jannu Soloman: Conceptualization, Methodology, Supervision, Review and Editing.
  Nagaraju Baydeti: Conceptualization, Methodology, Supervision, Review and Editing.

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